

		metal plate.
19	B	The electric fields at the middle point due to X and Y both act towards the right. Thus, the net electric field strength is equal to the sum of the component electric field strengths. $E_{\text{total}} = E_1 + (-E_2)$ $= 2 \times \frac{Q}{4\pi\epsilon_0 r^2}$ $= \frac{Q}{2\pi\epsilon_0 r^2}$ Note: Avoid putting the negative sign for the charge on Y. Follow the sign convention for the field lines. At the middle, the field lines due to X and Y are both positive (i.e. towards the right), so they add up.
20	D	As voltage across the lamp increases, its temperature increases, leading to more vigorously vibration of lattice ions. The rate of collision between free electrons and